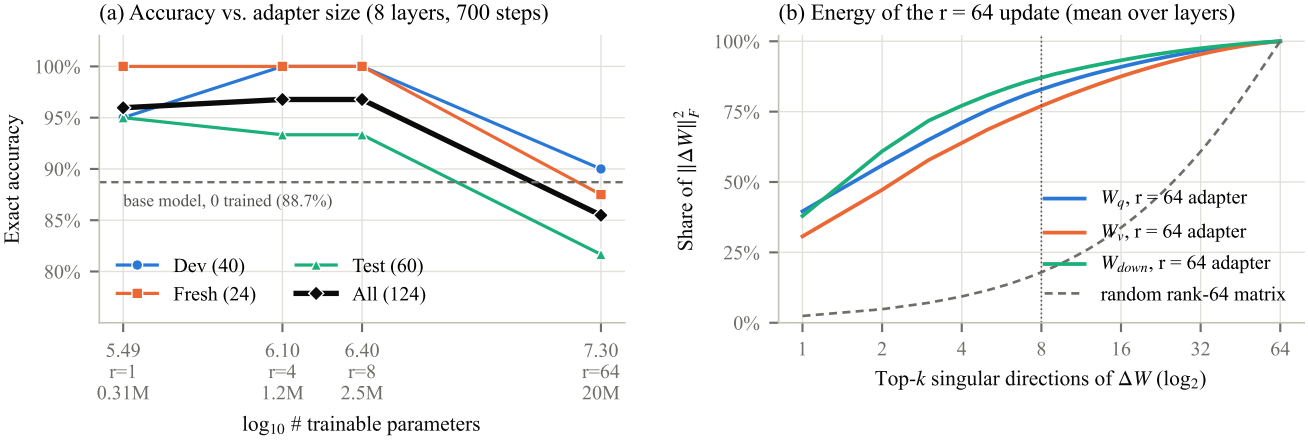
(a) Accuracy vs. adapter size (8 layers, 700 steps)
Exact accuracy
100%
95%
90%
85%
80%
base model, 0 trained (88.7%)
Dev (40)
Test (60)
Fresh (24)
All (124)
5.49
r=1
0.31M
6.10
r=4
1.2M
6.40
r=8
2.5M
7.30
r=64
20M
log10 # trainable parameters
(b) Energy of the r = 64 update (mean over layers)
Share of ||ΔW||²F
100%
75%
50%
25%
0%
Wq, r = 64 adapter
Wv, r = 64 adapter
Wdown, r = 64 adapter
random rank-64 matrix
1
2
4
8
16
32
64
Top-k singular directions of ΔW (log2)